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Opening and closing mechanism and leakage protection device

Abstract

Embodiments of the present disclosure provide an opening and closing mechanism and a leakage protection device. The opening and closing mechanism comprises: a stationary contact and a movable contact, which contacts the stationary contact at ON-position and separates from the stationary contact at OFF-position; a first magnetic assembly; and a second magnetic assembly that is integrally rotatable about an axis, the second magnetic assembly comprising: an electric conductor onto which the movable contact is mounted; and a magnetic body comprising at least one magnetic portion configured to rotate about the axis under a magnetic force from the first magnetic assembly, to correspondingly drive the movable contact to rotate between the ON-position and the OFF-position, wherein a distance from each of the at least one magnetic portion to the axis is smaller than a distance from the movable contact to the axis. In accordance with embodiments of the present disclosure, the transmission efficiency within the opening and closing mechanism may be boosted, and the movable and stationary contacts are separated faster and by a longer distance, so as to improve the performance of the opening and closing mechanism.

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Background/Summary

FIELD

(1) The present disclosure relates to the field of electrical equipment, and more specifically, to an opening and closing mechanism and a leakage protection device comprising the same.

BACKGROUND

(2) Leakage protection devices, such as Ground Fault Circuit Interrupter (GFCI) or Residual Current Device (RCD), can detect leakage and rapidly cut off the power supply when the leakage occurs, so as to protect safety of the personnel and the load. The leakage protection devices usually include an opening and closing mechanism for cutting off the power supply or breaking the

currents. The opening and closing mechanism may execute opening and closing operations. Once detecting leakage, the leakage protection device may indicate the opening and closing mechanism to perform an opening operation, to disconnect the load from the power supply. After the leakage is resolved, the opening and closing mechanism may execute a closing operation, allowing the power supply to continue to supply power to the load.

(3) Existing opening and closing mechanisms have various issues. For example, the opening speed of the opening and closing mechanism is slow, and the internal transmission structure of the opening and closing mechanism is complicated and low-efficient etc. The above issues impact performance and efficiency of the opening and closing mechanism.

SUMMARY

(4) To at least solve the above and other possible issues, embodiments of the present disclosure provide a new opening and closing mechanism and a leakage protection device comprising the opening and closing mechanism.

(5) In a first aspect of the present disclosure, there is provided an opening and closing mechanism, comprising a stationary contact and a movable contact, which contacts the stationary contact at ON-position and separates from the stationary contact at OFF-position; a first magnetic assembly; and a second magnetic assembly that is integrally rotatable about an axis, the second magnetic assembly comprising: an electric conductor onto which the movable contact is mounted; and a magnetic body comprising at least one magnetic portion configured to rotate about the axis under a magnetic force from the first magnetic assembly, to correspondingly drive the movable contact to rotate between the ON-position and the OFF-position, wherein a distance from each of the at least one magnetic portion to the axis is smaller than a distance from the movable contact to the axis.

(6) In the embodiments of the present disclosure, the integrated rotary structure drives the movable contact to move and rotation radius at the driving side of the rotary structure and the rotation radius at the side of the movable contact are appropriately configured. In such case, the transmission efficiency may be boosted, and the movable and stationary contacts are separated faster and by a longer distance, so as to improve the performance of the opening and closing mechanism.

(7) In some implementations of the present disclosure, the electric conductor and the magnetic body of the second magnetic assembly are formed into an integral component by injection molding. In this way, possible gaps and frictions in the transmission can be effectively eliminated, to further boost the transmission efficiency.

(8) In some implementations of the present disclosure, the at least one magnetic portion comprises a first magnetic portion and a second magnetic portion both having a first polarity, and a third magnetic portion and a fourth magnetic portion both having a second polarity opposite to the first polarity, and wherein when the movable contact is at the ON-position, the first magnetic portion and the third magnetic portion form a closed magnetic circuit with the first magnetic assembly, and when the movable contact is at the OFF-position, the second magnetic portion and the fourth magnetic portion form a closed magnetic circuit with the first magnetic assembly. In these implementations, a closed magnetic circuit is formed, which may reduce leakage of the magnetic flux and promote the efficiency for converting the electromagnetic energy to kinetic energy. In addition, the closed magnetic circuit may generate a greater electromagnetic force than the open magnetic circuit.

(9) In some implementations of the present disclosure, the first magnetic assembly comprises a magnetizer and an electromagnetic coil wound around the magnetizer, the electromagnetic coil being configured to introduce two currents in different directions, and wherein the first magnetic portion and the fourth magnetic portion are arranged adjacent to a first pole of the magnetizer, and the second magnetic portion and the third magnetic portion are arranged adjacent to a second pole of the magnetizer, the first pole and the second pole having opposite polarity. In the implementations, the interaction force between all magnetic portions and the first magnetic assembly is fully utilized and a closed magnetic circuit is also formed, to more rapidly drive the

movable contact to rotate.

(10) In some implementations of the present disclosure, the second magnetic assembly comprises a first support and a rotation shaft disposed along the axis, wherein the electric conductor and the magnetic body are fixed to the first support, and the first support is connected to the rotation shaft to rotate around the axis. In these implementations, when the electric conductor and the magnetic body are fixed by the first support to form an integrated component, the influence of the gaps and frictions between the components on the transmission is greatly reduced.

(11) In some implementations of the present disclosure, the first magnetic assembly comprises a second support, wherein the second support is fixed to the magnetizer and the rotation shaft is rotatably connected to the second support. In these implementations, the first magnetic assembly and the second magnetic assembly are arranged in a more compact manner, to downsize the overall volume of the opening and closing mechanism.

(12) In some implementations of the present disclosure, the magnetic body comprises a permanent magnet magnetically coupled to the at least one magnetic portion made of magnetically conductive materials. In the implementations, the magnetic body of the second magnetic assembly in need of rotation is formed via a simple, reliable and low-cost approach.

(13) In some implementations of the present disclosure, the electric conductor comprises an elastic copper sheet. These implementations may reduce the electromagnetic power required for rotating the movable contact in the breaking operation and downsize the volume of the magnetizer or iron core in the first magnetic assembly.

(14) In some implementations of the present disclosure, a U-shaped protrusion or a slot is formed on the elastic copper sheet. The U-shaped protrusion and slot may increase elasticity of the elastic copper sheet. In addition, due to the U-shaped protrusion, the elastic copper sheet gains a greater flexibility of the length and may be applied to a wider range. Moreover, the arrangement of the slot reduces the materials to be used, i.e., the costs are further lowered.

(15) In some implementations of the present disclosure, the opening and closing mechanism further comprises: an elastic piece disposed adjacent to the first magnetic portion, the elastic piece being configured to press the first magnetic portion towards the ON-position when the movable contact is at the OFF-position, and to separate from the first magnetic portion when the movable contact is at the ON-position. In this way, the electromagnetic power required for rotating the movable contact in the closing operation is reduced and the volume of the magnetizer or iron core in the first magnetic assembly is downsized.

(16) In a second aspect of the present disclosure, there is provided a leakage protection device, comprising: the opening and closing mechanism according to the first aspect. The advantageous effects obtained by the first aspect also apply to the second aspect.

(17) In some implementations of the present disclosure, the leakage protection device comprises: a control unit coupled to the first magnetic assembly and configured to switch on and off the opening and closing mechanism by controlling the first magnetic assembly, and a communication unit coupled to the control unit, the communication unit being configured to communicate with an external device. In these implementations, the operating staff may remotely operate the leakage protection device. Accordingly, the operating staff may safely and conveniently operate the leakage protection device.

(18) In a third aspect of the present disclosure, there is provided an electrical apparatus, comprising the opening and closing mechanism of the first aspect. The advantageous effects obtained by the first aspect also apply to the third aspect.

(19) This Summary is provided to introduce a selection of concepts in a simplified form that are further described below in the Detailed Description. This Summary is not intended to identify key features or essential features of the claimed subject matter, nor is it intended to be used to limit the scope of the claimed subject matter.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Through the following more detailed description of the example embodiments of the present disclosure with reference to the accompanying drawings, the above and other objectives, features, and advantages of the present disclosure will become more apparent, wherein the same reference sign usually refers to the same component in the example embodiments of the present disclosure:

(2) FIG. **1** illustrates a leakage protection device in accordance with embodiments of the present disclosure;

(3) FIG. **2** illustrates a section view of the opening and closing mechanism in accordance with embodiments of the present disclosure;

(4) FIG. **3**A illustrates a perspective view of the ON-state opening and closing mechanism in accordance with embodiments of the present disclosure;

(5) FIG. **3**B illustrates a perspective view of the OFF-state opening and closing mechanism in accordance with embodiments of the present disclosure;

(6) FIG. **4** illustrates a perspective view of the second magnetic assembly in accordance with embodiments of the present disclosure;

(7) FIG. **5**A illustrates a front view of the second magnetic assembly in accordance with embodiments of the present disclosure;

(8) FIG. **5**B illustrates a section view of the second magnetic assembly in accordance with embodiments of the present disclosure;

(9) FIG. **6**A illustrates a perspective view of the opening and closing mechanism in accordance with alternative embodiments of the present disclosure;

(10) FIG. **6**B illustrates a perspective view of the electric conductor in accordance with alternative embodiments of the present disclosure;

(11) FIG. **7** illustrates an analysis chart of magnetic field simulation of the magnetic body of the first magnetic assembly and the second magnetic assembly;

(12) FIG. **8** illustrates a perspective view of the second support in the first magnetic assembly in accordance with embodiments of the present disclosure;

(13) FIG. **9**A illustrates a front view of the OFF-state opening and closing mechanism in accordance with alternative embodiments of the present disclosure;

(14) FIG. **9**B illustrates a front view of the ON-state opening and closing mechanism in accordance with alternative embodiments of the present disclosure;

(15) FIG. **10** illustrates a schematic diagram of the leakage protection device communicating with an external device in accordance with embodiments of the present disclosure.

DETAILED DESCRIPTION OF EMBODIMENTS

(16) The embodiments of the present disclosure will be described in more details with reference to the drawings. Although the drawings illustrate the embodiments of the present disclosure, it should be appreciated that the present disclosure can be implemented in various manners and should not be limited to the embodiments explained herein. On the contrary, the embodiments are provided to make the present disclosure more thorough and complete and to fully convey the scope of the present disclosure to those skilled in the art. Those skilled in the art may obtain an alternative technical solution from the following description without deviating from the spirit and the protection scope of the present disclosure.

(17) As used herein, the term “includes” and its variants are to be read as open-ended terms that mean “includes, but is not limited to.” The term “or” is to be read as “and/or” unless the context clearly indicates otherwise. The term “based on” is to be read as “based at least in part on.” The terms “one example embodiment” and “one embodiment” are to be read as “at least one example embodiment.” The following text also can include other explicit and implicit definitions.

(18) Embodiments of the present disclosure provide a novel opening and closing mechanism and a leakage protection device comprising the opening and closing mechanism. Within the opening and closing mechanism or the leakage protection device, a magnetic assembly that drives a movable contact to move is an integrated rotary structure, to effect an efficient gapless transmission with small friction. In addition, greater separation velocity and wider distance may be obtained between a movable contact and a stationary contact by appropriately setting radius of rotation at a driving side and radius rotation at a driven side. This improved opening and closing mechanism can effectively promote efficiency of breaking and connecting operations and the overall performance.
(19) FIG. **1** illustrates a leakage protection device **1000** in accordance with embodiments of the present disclosure. As an example, the leakage protection device **100** may be Ground Fault Circuit Interrupter (GFCI) or Residual Current Device (RCD). However, it is to be understood that the leakage protection device **100** is not limited to this and instead may be any other devices that fulfill leakage protection purpose.
(20) FIG. **2** illustrates a section view of the opening and closing mechanism **100** in accordance with embodiments of the present disclosure; FIG. **3**A illustrates a perspective view of the ON-state opening and closing mechanism **100** in accordance with embodiments of the present disclosure; and FIG. **3**B illustrates a perspective view of the OFF-state opening and closing mechanism **100** in accordance with embodiments of the present disclosure. The opening and closing mechanism **100** may be disposed in the leakage protection device **1000** in FIG. **1**. However, the present disclosure is not restricted to this. The opening and closing mechanism **100** may also be disposed in any other devices that fulfill the protection purpose by engaging or disengaging the contact. For example, the opening and closing mechanism **100** may be arranged in electricity meter, small-scale circuit breaker or Miniature Circuit Breaker (MCB), Molded Case Circuit Breaker (MCCB), socket or other types of electromagnetic switches (e.g., relay).
(21) In accordance with embodiments of the present disclosure, the opening and closing mechanism **100** may include a stationary contact **110** and a movable contact **120**, which contacts the stationary contact **110** at an ON-position (as shown in FIG. **3**A) and is separated from the stationary contact **110** at an OFF-position (as shown in FIG. **3**B). As an example, the opening and closing mechanism **100** may be connected in a main circuit between a power supply and a load. The stationary contact **110** may be used for a circuit in the main circuit connected to the power supply. Correspondingly, the movable contact **120** may be used for a circuit in the main circuit connected to the load, and vice versa. The movable contact **120** is capable of performing a moving operation, so that the movable contact **120** may move between the ON-position where it makes contact with the stationary contact **110** and the OFF-position where it separates from the stationary contact **110**, so as to connect or disconnect the main circuit as a function of the indication.
(22) In accordance with embodiments of the present disclosure, the opening and closing mechanism **100** may include a first magnetic assembly **130** and a second magnetic assembly **140**. As an example, the first magnetic assembly **130** and the second magnetic assembly **140** may be independent of each other. The first magnetic assembly **130** and the second magnetic assembly **140** may include an essential element for producing or transmitting magnetic flux, e.g., a permanent magnet comprised of magnetized hard magnetic materials, an iron core or magnetizer consisting of soft magnetic materials, and a coil wound around the iron core or the magnetizer etc.
(23) An exemplary structure of the second magnetic assembly **140** is further described below with reference to FIGS. **4**, **5**A and **5**B on the basis of FIGS. **2**, **3**A and **3**B. FIG. **4** illustrates a perspective view of the second magnetic assembly **140** in accordance with embodiments of the present disclosure; FIG. **5**A illustrates a front view of the second magnetic assembly **140** and FIG. **5**B illustrates a section view of the second magnetic assembly **140**. In accordance with embodiments of the present disclosure, the second magnetic assembly **140** can integrally rotate about X-axis. The second magnetic assembly **140** includes an electric conductor **141**, on to which the movable contact **120** is mounted. As an example, the second magnetic assembly **140** as a whole

can rotate about the X-axis, to drive the rotation of the electric conductor **141**. In comparison to other transmission structures, e.g., multi-level transmission structures or rotational-to-linear motion transmission structures, the integrated rotary structure can more efficiently transmit power and avoid frictions and gaps between various components of the transmission structure. As the movable contact **120** is mounted on the electric conductor **141**, the rotation of the second magnetic assembly **140** will correspondingly drive the movable contact **120** on the electric conductor **141** to rotate. For example, the movable contact **120** may be mounted at one end of the electric conductor **141**, and the other end of the electric conductor **141** is connected to the main circuit. Accordingly, in case of the movable contact **120** contacting the stationary contact **110**, the electric conductor **141** may transmit supply power from the stationary contact **110** to post-stage loads, or to post-stage loads at the side of the stationary contact.
(24) In accordance with embodiments of the present disclosure, the second magnetic assembly **140** further include a magnetic body **142**, wherein the magnetic body **142** consists of at least one magnetic portion **1421**, **1422**, **1423**, **1424** configured to rotate about the X-axis under the magnetic action from the first magnetic assembly **130**, to correspondingly drive the movable contact **120** to rotate between the ON-position and the OFF-position. A distance R**1** from each of the at least one magnetic portion **1421**, **1422**, **1423**, **1424** to the X-axis is smaller than a distance R**2** from the movable contact **120** to the X-axis. The number of magnetic portions is not restricted to the amount shown in the drawings. Instead, any suitable numbers of magnetic portions may be provided as required. The magnetic portion **1421**, **1422**, **1423**, **1424** may be affected by the magnetic force of the first magnetic assembly **130** and rotate about the X-axis under the action of the magnetic force, thereby driving the movable contact **120** to rotate about the X-axis. If a distance R**1** from any of the magnetic portion **1421**, **1422**, **1423**, **1424** to the X-axis (only distance R**1** from the magnetic portion **1422** to the X-axis is shown as an example) is smaller than a distance from the movable contact **120** to the X-axis, it means that a radius of the rotating track of any magnetic portion is smaller than a radius of the rotating track of the movable contact **120**. As a result, a greater linear velocity is generated to separate the movable contact **120** from the stationary contact **110**, and the rotation of the movable contact **120** may span a longer distance. For example, as shown in FIG. **3**B, when the magnetic portion rotates for a distance D**1**, the stationary contact **110** and the movable contact **120** may be spaced apart for a longer distance D**2**. On account of a longer breaking distance and a greater breaking velocity, the stationary contact **110** is quickly and efficiently separated from the movable contact **120** and arcs that may possibly be generated between the stationary contact **110** and the movable contact are extinguished, thereby cutting off the main circuit and further improving the shutoff performance of the opening and closing mechanism.
(25) In some embodiments of the present disclosure, the electric conductor **141** and the magnetic body **142** of the second magnetic assembly **140** are formed into an integrated component by injection molding. As an example, the electric conductor **141** and the magnetic body **142** are formed into one piece by injection molding. Accordingly, the electric conductor **141** and the magnetic body **142** are stably integrated by injection molding materials, to effectively eliminate possible gaps and frictions in the transmission. These gaps and frictions are adverse factors that lower the transmission efficiency. When the magnetic portion of the magnetic body **142** is driven by the magnetic force to rotate, the driving force may be more efficiently transmitted to the movable contact **120**.
(26) In some embodiments of the present disclosure, for example in FIGS. **4** and **5**B, the magnetic body **142** includes a permanent magnet **1425** magnetically coupled to at least one magnetic portion **1421**, **1422**, **1423**, **1424** made of magnetically conductive materials. As an example, each of the at least one magnetic portion **1421**, **1422**, **1423**, **1424** may be a part of a sheet comprised of magnetically conductive materials, and the magnetic body **142** may include one or more sheets, e.g., two sheets. For example, the two sheets may be disposed at both sides of the permanent magnet **1425**, i.e., S-pole side and N-pole side of the permanent magnet **1425**. Therefore, the

magnetic flux generated by the permanent magnet **1425** may penetrate through the at least one magnetic portion **1421**, **1422**, **1423**, **1424**, to produce a magnetic force between the at least one magnetic portion **1421**, **1422**, **1423**, **1424** and the first magnetic assembly **130**. It is to be understood that the at least one magnetic portion **1421**, **1422**, **1423**, **1424** of the magnetic body **142** may also be composed of permanent magnets directly, or iron core or electromagnetic coil in replacement of the permanent magnet **1425**. It is noted that the use of the permanent magnet **1425** will bring a big edge over other approaches. Specifically, if each magnetic portion is made of permanent magnets, it is more difficult to manufacture the second magnetic assembly **140**, which accordingly increases the costs of production. In case that the iron core and the electromagnetic coil are employed, it is also required to introduce some additional assemblies, such as brush or conductive ring, to enable the rotation of the second magnetic assembly **140**. In such case, the manufacturing process becomes more complicated and the reliability decreases. Hence, it is a simple, low-cost and reliable approach to use the permanent magnet **1425** as the magnetic body **142** of the second magnetic assembly **140**.

(27) In some embodiments of the present disclosure, the electric conductor **141** of the second magnetic assembly **140** may include an elastic copper sheet, which may conduct currents at a quite low loss. In addition, the sufficient strength of the elastic copper sheet may help fixing the movable contact **120** and driving it to rotate. The elastic copper sheet also has an advantage in driving the movable contact **120** to rotate. To be specific, when the movable contact **120** is at the ON-position and it is required to separate it from the stationary contact **110**, an elastic force of the elastic copper sheet may provide an auxiliary torque for the rotation of the movable contact **120**, which may reduce the electromagnetic power required for rotating the movable contact in the breaking operation and downsize the volume of the magnetizer or iron core in the first magnetic assembly **130**. In some embodiments of the present disclosure, a U-shaped protrusion **1411** is formed on the elastic copper sheet. As an example, the U-shaped protrusion may be positioned between an anchor point for fixing the elastic copper sheet and the movable contact **120**, to increase elasticity of the elastic copper sheet. When a length of the elastic copper sheet is not suitable for the opening and closing mechanism **100**, the U-shaped protrusion may adjust the length of the elastic copper sheet to gain a greater flexibility. FIGS. **6**A and **6**B respectively illustrate perspective views of the opening and closing mechanism **100** and the electric conductor **141** in accordance with alternative embodiments of the present disclosure. As shown in FIGS. **6**A and **6**B, in some embodiments of the present disclosure, a slot **1412** is formed on the elastic copper sheet serving as the electric conductor **141**. As an example, the slot **1412** may be positioned on an overhanging part of the elastic copper sheet and present in an elongated shape, wherein the slot **1412** may be substantially positioned at the center of a width direction of the elastic copper sheet and extend along a length direction of the elastic copper sheet. It is to be appreciated that the position and the shape of the slot **1412** are not restricted to the above features. Instead, the slot may be formed at other suitable positions or into other appropriate shapes as needed. Advantageously, the slot **1412** may enhance elasticity of the elastic copper sheet and reduce the weight and the materials of the elastic copper. This further lowers the manufacturing costs.

(28) In some embodiments of the present disclosure, the at least one magnetic portion **1421**, **1422**, **1423**, **1424** includes a first magnetic portion **1421** and a second magnetic portion both having a first polarity, and a third magnetic portion **1423** and a fourth magnetic portion **1424** both having a second polarity opposite to the first polarity. When the movable contact **120** is at the ON-position, the first magnetic portion **1421** and the third magnetic portion **1423** form a closed magnetic circuit with the first magnetic assembly **130**. If the movable contact **120** is at the OFF-position, the second magnetic portion **1422** and the fourth magnetic portion **1424** form a closed magnetic circuit with the first magnetic assembly **130**. As an example, the magnetic body **142** may be formed into a shape similar to "I-shape", and the at least one magnetic portion **1421**, **1422**, **1423**, **1424** is respectively located at four corners of the I-shape. For example, as illustrated in FIG. **5**B, the first

magnetic portion **1421** and the second magnetic portion **1422** exhibit S polarity, while the third magnetic portion **1423** and the fourth magnetic portion **1424** exhibit N polarity. Accordingly, when the first magnetic portion **1421** having S polarity and the third magnetic portion **1423** having N polarity are respectively driven by a magnetic attraction force exerted by the first magnetic assembly **130** to come near at or contact the first magnetic assembly **130** (i.e., the case shown in FIGS. **2** and **3**A when the movable contact **120** is at the ON-position), the magnetic flux may pass through the first magnetic portion **1421**, the third magnetic portion **1423** and the first assembly **130** to form a closed magnetic circuit. Similarly, when the second magnetic portion **1422** having S polarity and the fourth magnetic portion **1424** having N polarity are respectively driven by a magnetic attraction force exerted by the first magnetic assembly **130** to come near at or contact the first magnetic assembly **130** (i.e., the case shown in FIG. **3**B when the movable contact **120** is at the OFF-position), the magnetic flux may pass through the second magnetic portion **1422**, the fourth magnetic portion **1424** and the first assembly **130** to form a closed magnetic circuit. It is to be understood that the first magnetic portion **1421** and the second magnetic portion **1422** may have N polarity, while the third magnetic portion **1423** and the fourth magnetic portion **1424** may exhibit S polarity in other embodiments. In such case, embodiments of the present disclose may still be implemented. During the magnetic interaction between the first magnetic assembly **130** and the second magnetic assembly **140**, if it is ensured that a closed magnetic circuit is formed, leakage of the magnetic flux may be reduced, so as to promote the efficiency for converting the electromagnetic energy to kinetic energy. The closed magnetic circuit may generate a greater electromagnetic force than the open magnetic circuit. In such case, a sufficient power may be provided to drive the magnetic portion of the magnetic body **142** to rotate, thereby separating the movable contact and the stationary contact. Particularly, when the distance from the magnetic portion **1421**, **1422**, **1423**, **1424** to the X-axis is smaller than the distance from the movable contact **120** to the X-axis, the movable contact is rotated only when the magnetic force tolerated by the magnetic portion is larger than the force that prevents the movable contact **120** from moving. Therefore, a greater electromagnetic force may be provided by the closed magnetic circuit structure to move the movable contact **120** to implement shutoff operation and breaking operation.

(29) In some embodiments of the present disclosure, the first magnetic assembly **130**, as shown in FIG. **2**, includes a magnetizer **131** and an electromagnetic coil **132** wound around the magnetizer **131**, the electromagnetic coil **132** being configured to introduce two currents in different directions. The first magnetic portion **1421** and the fourth magnetic portion **1424** are disposed as a first pole **1311** adjacent to the magnetizer **131**, and the second magnetic portion **1422** and the third magnetic portion **1423** are arranged as a second pole **1312** adjacent to the magnetizer **131**. The first pole **1311** and the second pole **1312** have opposite polarity. As an example, forward DC currents and reverse DC currents may be passed into the electromagnetic coil **132**, to generate two magnetic fields having different magnetic flux directions in the magnetizer **131**. For example, the electromagnetic coil **132** may include a single winding, into which forward currents or inverse currents may be introduced by appropriate control operations. Alternatively, the electromagnetic coil **132** may include forward winding and reverse winding, wherein forward currents are passed into the forward winding and reverse currents are introduced into the reverse winding. When forward currents are introduced into the electromagnetic coil **132**, the first pole **1311** of the magnetizer **131** adjacent to the first magnetic portion **1421** and the fourth magnetic portion **1424** has N polarity, i.e., the magnetic flux outflows from the first pole **1311**, goes into the first magnetic portion **1421** and reaches the S pole of the permanent magnet **1425**. Accordingly, a magnetic attraction force exists between the first magnetic portion **1421** and the first pole **1311**, and a magnetic repulsion force is present between the fourth magnetic portion **1424** and the first pole **1311**. The first magnetic portion **1421** and the fourth magnetic portion **1424** rotate clockwise. Meanwhile, the second pole **1312** of the magnetizer **131** adjacent to the second magnetic portion **1422** and the third magnetic portion **1423** exhibits S polarity, i.e., the magnetic flux starts from the

N pole of the permanent magnet **1425**, outflows from the third magnetic portion **1423** and enters the second pole **1312**. Accordingly, a magnetic repulsion force is present between the second magnetic portion **1422** and the second pole **1312**, and a magnetic attraction force exists between the third magnetic portion **1423** and the second pole **1312**. The second magnetic portion **1422** and the third magnetic portion **1423** rotate clockwise. On the other hand, when reverse currents are introduced into the electromagnetic coil **132**, the first pole **1311** of the magnetizer **131** adjacent to the first magnetic portion **1421** and the fourth magnetic portion **1424** has S polarity, i.e., the magnetic flux starts from the N pole of the permanent magnet **1425**, outflows from the fourth magnetic portion **1424** and goes into the first pole **1312**. Accordingly, a magnetic repulsion force exists between the first magnetic portion **1421** and the first pole **1311**, and a magnetic attraction force is present between the fourth magnetic portion **1424** and the first pole **1311**. The first magnetic portion **1421** and the fourth magnetic portion **1424** rotate counter-clockwise. Meanwhile, the second pole **1312** of the magnetizer **131** adjacent to the second magnetic portion **1422** and the third magnetic portion **1423** exhibits N polarity, i.e., the magnetic flux outflows from the second pole **1312**, goes into the second magnetic portion **1422** and reaches the S pole of the permanent magnet **1425**. Accordingly, a magnetic attraction force is present between the second magnetic portion **1422** and the second pole **1312**, and a magnetic repulsion force exists between the third magnetic portion **1423** and the second pole **1312**. The second magnetic portion **1422** and the third magnetic portion **1423** rotate counter-clockwise.

(30) In this approach, by controlling the current direction of the electromagnetic coil, the four magnetic portions **1421**, **1422**, **1423** and **1423** may be simultaneously driven to rotate clockwise, so as to drive the movable contact **120** to rotate to the ON-position where it contacts the stationary contact **110**. Alternatively, the four magnetic portions **1421**, **1422**, **1423** and **1423** may be simultaneously driven to rotate counter-clockwise, so as to drive the movable contact **120** to rotate to the OFF-position where it leaves from the stationary contact **110**. In the above approach, the interaction force between all magnetic portions of the magnetic body **142** and the magnetic assembly **130** is fully utilized, to rapidly and effectively drive the second magnetic assembly **140** to rotate.

(31) Moreover, FIG. **7** illustrates an analysis chart of magnetic field simulation of the magnetic body **142** of the first magnetic assembly **130** and the second magnetic assembly **140**. It is known from the chart that the two magnetic portions at diagonal positions of the magnetic body **142**, the permanent magnet **1425** and the magnetizer **131** (including its body and two poles **1311** and **1312**) form a closed magnetic circuit, to take full advantage of the electromagnetic force and the permanent magnetic force.

(32) In some embodiments of the present disclosure, as shown in FIGS. **3**A and **4**A, the second magnetic assembly **140** includes a first support **145** and a rotation shaft **146** disposed along the X-axis. The electric conductor **141** and the magnetic body **142** are fixed on the first support **145**, which first support **145** is connected to the rotation shaft **146** to rotate about the X-axis. As an example, with reference to FIG. **4**, the first support **145** is formed on the electric conductor **141** and the magnetic body **142** by injection molding or molding, and the rotation shaft **146** may be formed integrally with the first support **145**, to minimize the influence of the gap and the friction between the components on the rotation transmission. Alternatively, the electric conductor **141**, the magnetic body **142**, the first support **145** and the rotation shaft **146** also may be connected and fixed by other suitable ways as long as they are integrated in a tight and basically gap-free manner.

(33) In some embodiments of the present disclosure, the first magnetic assembly **130** includes a second support **135** as shown in FIG. **3**A. The second support **135** is fixed to the magnetizer **131** and the rotation shaft **146** is rotatably connected to the second support **135**. As an example, FIG. **8** illustrates a perspective view of the second support **135** in accordance with embodiments of the present disclosure. According to FIG. **8**, a rotation shaft hole **1351** and a mounting hole **1352** are formed on the second support **135**. The second support **135** may be mounted to the first pole **1311**

and the second pole **1312** of the magnetizer **131** by the mating between the fastening elements and the mounting hole **1352**. Alternatively, the second support **135** also may be mounted at other suitable positions, e.g., housing of the opening and closing mechanism **100**, as long as the second support **135** can be fixed relative to the first magnetic assembly **130** and adjacent to the second magnetic assembly **140**. However, the second support **135** is preferably fixed to the magnetizer **131**, such that the two magnetic assemblies of the opening and closing mechanism **100** are more compact and have small volumes. In addition, the rotation shaft **146** may be mounted in the rotation shaft hole **1351** of the second support **135**. In such case, the second support **135** may support the second magnetic assembly **140** in a vertical direction, and the second magnetic assembly **140** may rotate as a whole with respect to the second support **135** and the first magnetic assembly **130**.

(34) FIG. **9**A illustrates a front view of the OFF-state opening and closing mechanism **100** in accordance with alternative embodiments of the present disclosure, and FIG. **9**B illustrates a front view of the ON-state opening and closing mechanism **100** in accordance with alternative embodiments of the present disclosure. In some embodiments of the present disclosure, the opening and closing mechanism **100** also may include an elastic piece **150** arranged close to the first magnetic portion **1421**. The elastic piece **150** is configured to press the first magnetic portion **1421** towards the ON-position when the movable contact **120** is located at the OFF-position, and to separate the first magnetic portion **1421** when the movable contact **120** is located at the ON-position. As an example, one end of the elastic piece **150**, for example, may be connected and fixed to the housing of the opening and closing mechanism **100** or other components fixed relative to the housing. As shown in FIG. **9**A, when the opening and closing mechanism **100** is in the OFF-state (i.e., the movable contact **120** at the OFF-position), the elastic piece **150** close to the first magnetic portion **1421** will press the first magnetic portion **1421**. In this way, when the movable contact **120** needs to rotate from the OFF-position to the ON-position with respect to the stationary contact **110**, the elastic piece **150** may provide an auxiliary torque to assist the movable contact **120** completing the rotation. As a result, the electromagnetic power required for rotating the movable contact is reduced and the volume of the electric conductor or the iron core of the first magnetic assembly **130** is downsized. While the movable contact **120** is rotating from the OFF-position to the ON-position, the elastic piece **150** is separated from the first magnetic portion **1421**. For example, when the movable contact **120** is at one-third of the rotation course from the OFF-position to the ON-position, the elastic piece **150** is completely separated from the first magnetic portion **1421**. Therefore, as demonstrated by FIG. **9**B, when the opening and closing mechanism **100** is in the ON-state (i.e., the movable contact **120** is at the ON-position), the elastic piece **150** is disengaged from the contact with the first magnetic portion **1421** and kept at a distance from the first magnetic portion **1421**. Consequently, an error connection of the opening and closing mechanism **100** caused by elastic force exerted by the elastic piece **150** is avoided while an auxiliary rotation torque is still provided by the elastic piece **150**. In FIGS. **9**A and **9**B, the elastic piece **150** is illustrated as a coil spring as an example. However, implementations of the elastic piece **150** are not restricted to this. Instead, other types of elastic components are also feasible as long as they can fulfill bias, such as leaf spring or rubber-made elastomer etc.

(35) In the embodiments of the present disclosure, owing to the integrated rotary structure and suitable configurations of rotation radius of the magnetic portions and the movable contact, the transmission efficiency may be boosted, and the movable and stationary contacts are separated faster and by a longer distance, so as to improve the performance of the opening and closing mechanism.

(36) FIG. **10** illustrates a schematic diagram of the leakage protection device **1000** communicating with an external device **2000** in accordance with embodiments of the present disclosure. In some embodiments of the present disclosure, the leakage protection device **1000** may include a control unit **200** in addition to the opening and closing mechanism **100** as shown in FIG. **10**. The control

unit **200** is coupled to the first magnetic assembly **130** of the opening and closing mechanism **100** and configured to control ON and OFF of the opening and closing mechanism **100** via the first magnetic assembly **130**. As an example, the control unit **200** may include suitable power supply circuits and control circuits to control the first magnetic assembly **130**. For example, the opening and closing mechanism **100** is switched on and off by controlling directions and magnitudes of the currents in the electromagnetic coil **132** of the first magnetic assembly **130**.

(37) Furthermore, the leakage protection device **1000** also may include a communication unit **300** coupled to the control unit **200** and configured to communication with an external device **2000**. As an example, the external device **2000**, such as smartphone, tablet computer, notebook computer and desktop computer etc., may communicate with a communication unit **300** of the leakage protection device **100** in a wired or wireless way, so as to send control instructions to the communication unit **300**. The communication unit **300** may transmit the received control instructions to the control unit **200**, enabling the control unit **200** to connect and disconnect the opening and closing mechanism **100** in accordance with the control instructions from the external device **200**. Besides, the communication unit **300** also may send state information of the opening and closing mechanism **100** to the external device **2000**, which is further provided to operating staff. In this way, the operating staff may remotely operate the leakage protection device **1000** via the external device **2000**, to switch on or off the opening and closing mechanism **100**. As such, the operating staff may conveniently control the leakage protection device **1000**. It is to be understood that in addition to remote operation, the leakage protection device **1000** also may be operated locally to directly connect or disconnect the opening and closing mechanism **100**. Here, location operation, for example, includes manual or hand operation, or operation via local electric control components.

(38) In accordance with a further embodiment of the present disclosure, there is provided an electrical apparatus, which may include the opening and closing mechanism **100**. As an example, the electrical apparatus may be electricity meter, small-scale circuit breaker or Miniature Circuit Breaker (MCB), Molded Case Circuit Breaker (MCCB), socket or electromagnetic switches (e.g., relay). With the help of the opening and closing mechanism **100**, the electrical apparatus can connect and disconnect electrical lines as required. It is to be appreciated that the electrical apparatus is not limited to this. Instead, any electrical devices that can fulfill the particular function via closing or opening operations are feasible.

(39) Through the above description and teachings provided in the related drawings, many modifications and other implementations of the present disclosure disclosed herein will be conceived by those skilled in the field related to the present disclosure. It is to be understood that the implementations of the present disclosure are not limited to the specific implementations disclosed herein, and the modifications and other implementations are included within the scope of the present disclosure. Besides, although the example implementations have been described in the context of some example combinations of the components and/or functions with reference to the related drawings, it should be recognized that different combinations of the components and/or functions may be provided by alternative implementations without deviating from the scope of the present disclosure. As far as this is concerned, other combinations of components and/or functions distinct from the above clearly described ones are also expected to fall within the scope of the present disclosure. Although specific terms are used here, they only convey generic and descriptive meanings and are not intended as restrictions.

Claims

1. An opening and closing mechanism, comprising: a stationary contact and a movable contact, which contacts the stationary contact at ON-position and separates from the stationary contact at OFF-position; a first magnetic assembly; and a second magnetic assembly that is integrally rotatable about an axis, the second magnetic assembly comprising: an electric conductor onto

which the movable contact is mounted; and a magnetic body comprising at least one magnetic portion configured to rotate about the axis under a magnetic force from the first magnetic assembly, to correspondingly drive the movable contact to rotate between the ON-position and the OFF-position, wherein a distance from each of the at least one magnetic portion to the axis is smaller than a distance from the movable contact to the axis.

2. The opening and closing mechanism of claim 1, wherein the electric conductor and the magnetic body of the second magnetic assembly are formed into an integral component by injection molding.

3. The opening and closing mechanism of claim 1, wherein the at least one magnetic portion comprises a first magnetic portion and a second magnetic portion both having a first polarity, and a third magnetic portion and a fourth magnetic portion both having a second polarity opposite to the first polarity, and wherein when the movable contact is at the ON-position, the first magnetic portion and the third magnetic portion form a closed magnetic circuit with the first magnetic assembly, and when the movable contact is at the OFF-position, the second magnetic portion and the fourth magnetic portion form a closed magnetic circuit with the first magnetic assembly.

4. The opening and closing mechanism of claim 3, wherein the first magnetic assembly comprises a magnetizer and an electromagnetic coil wound around the magnetizer, the electromagnetic coil being configured to introduce two currents in different directions, and wherein the first magnetic portion and the fourth magnetic portion are arranged adjacent to a first pole of the magnetizer, and the second magnetic portion and the third magnetic portion are arranged adjacent to a second pole of the magnetizer, the first pole and the second pole having opposite polarity.

5. The opening and closing mechanism of claim 4, wherein the second magnetic assembly comprises a first support and a rotation shaft disposed along the axis, wherein the electric conductor and the magnetic body are fixed to the first support, and the first support is connected to the rotation shaft to rotate around the axis.

6. The opening and closing mechanism of claim 5, wherein the first magnetic assembly comprises a second support, wherein the second support is fixed to the magnetizer and the rotation shaft is rotatably connected to the second support.

7. The opening and closing mechanism of claim 1, wherein the magnetic body comprises a permanent magnet magnetically coupled to the at least one magnetic portion made of magnetically conductive materials.

8. The opening and closing mechanism of claim 1, wherein the electric conductor comprises an elastic copper sheet.

9. The opening and closing mechanism of claim 8, wherein a U-shaped protrusion or a slot is formed on the elastic copper sheet.

10. The opening and closing mechanism of claim 3, further comprising: an elastic piece disposed adjacent to the first magnetic portion, the elastic piece being configured to press the first magnetic portion towards the ON-position when the movable contact is at the OFF-position, and to separate from the first magnetic portion when the movable contact is at the ON-position.

11. A leakage protection device, comprising: the opening and closing mechanism according to claim 1.

12. The leakage protection device of claim 11, further comprising: a control unit coupled to the first magnetic assembly and configured to switch on and off the opening and closing mechanism by controlling the first magnetic assembly, and a communication unit coupled to the control unit, the communication unit being configured to communicate with an external device.

13. An electrical apparatus, comprising the opening and closing mechanism according to claim 1.